

SuperApp

A full pipeline Data Cleaning & Exploratory Data Analysis Application

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Problem

- Time-consuming converting invoice into usable excel dataset
- Highly repetitive manual data cleaning and naming/grouping convention
- No specific guideline for naming/grouping convention

Solution

- Seamless full pipeline Invoice Converter → Data Pre-processing → Data Visualization
- A 3-in-1 simple, simple & intuitive application usable by everyone regardless of their technical skill
- 2 reports output tailored for management's needs (Yearly Overview Report & Monthly Detailed Report)
- Reducing manual data entry which can save hours of work

Scope of Project

- Limited to WMG's Internal use only

Introducing: SuperApp

A seamless full pipeline 3-in-1 simple yet intuitive python-native Application tailored for the needs of WMG's Management and Data Analysis

The application is highly tailored to fulfill the requirements and makes it easy for everyone including non-technical person to use it.



Key features of SuperApp

Automated Invoice Converter

The invoices generated and extracted from SQL Accounting are still in the 'raw' format, hence it needs to be converted to usable & workable excel dataset

Seamless Data Cleaning and Grouping

The 2nd part of the application handles the basic data cleaning and outlet grouping, It uses mapping.csv to group each outlet to their belonging group. You can also add, edit, and delete the new or existing outlet names.

Auto Exploratory Data Analysis

The program will generate both overview and detailed visualization. User will be able to choose which report to export (Yearly Overview Report / Monthly Detailed Report)

Limitations and Suggestions

SuperApp is in its earliest version. In the event that the management requires update of the application, future developers need to edit/update the source code, and later package it again using PyInstaller.

SuperApp is limited to current/existing dataset analysis, suppose the management wants to see data forecasting, future developers are free to add forecasting function in the source code

SuperApp is wrapped in Tkinter GUI for simplicity and ease of use, suppose in the future you want to make it “fancier” feel free to overhaul the GUI

SuperApp cannot choose/differentiate LKA&IKA, it's solely depends on the input dataset

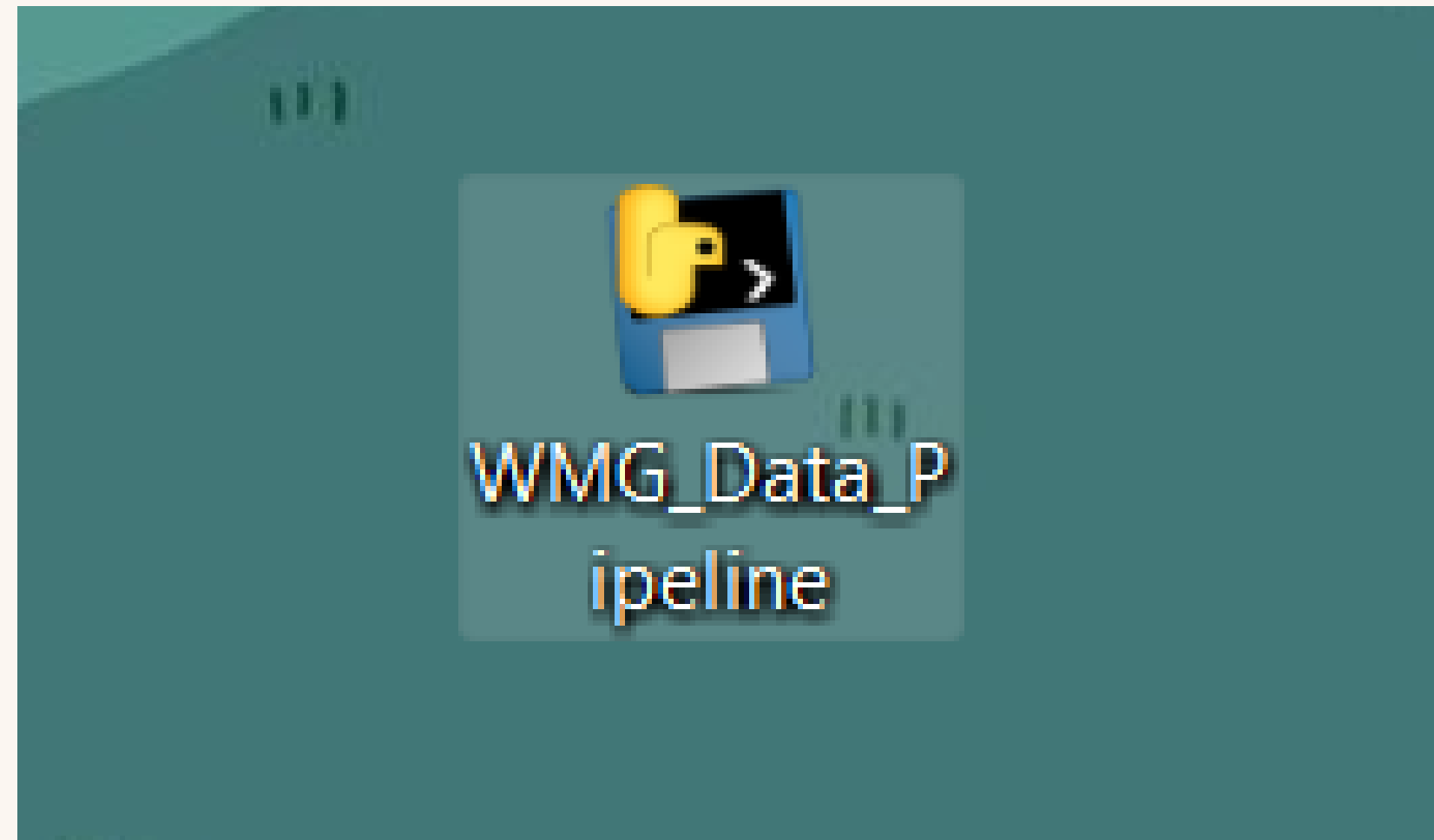
SuperApp is limited to WMG's Internal use only, as the source code include sample report using WMG's real sales data

Thank you & All the best !

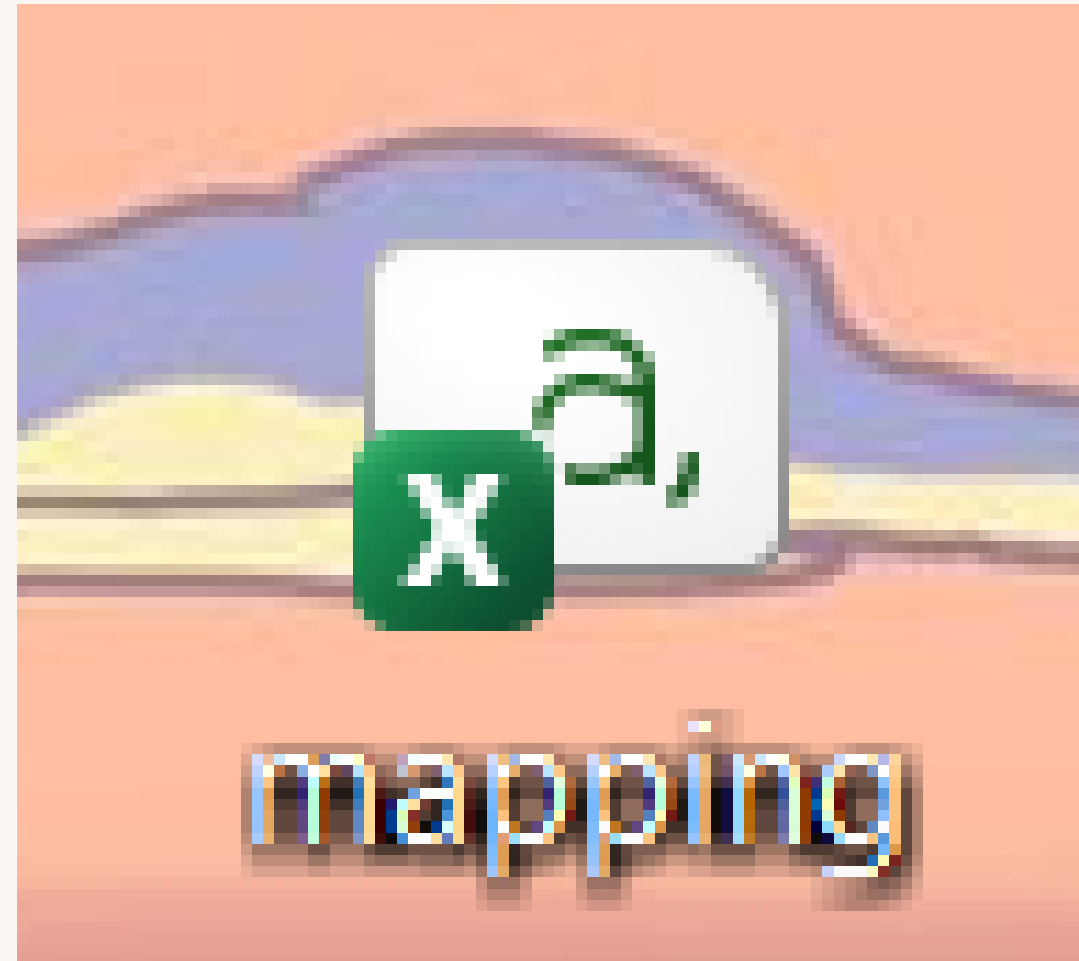


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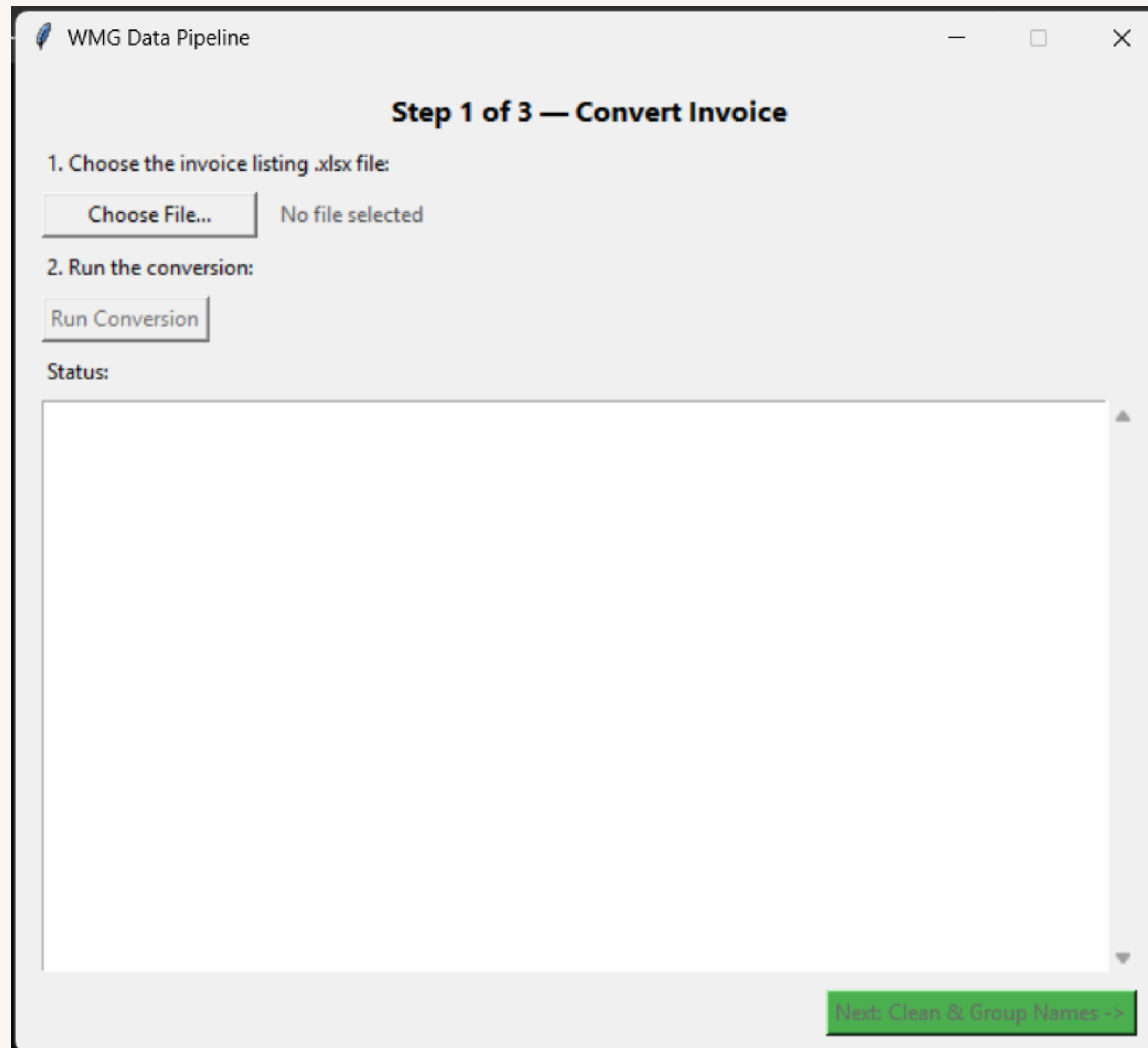
Appendix



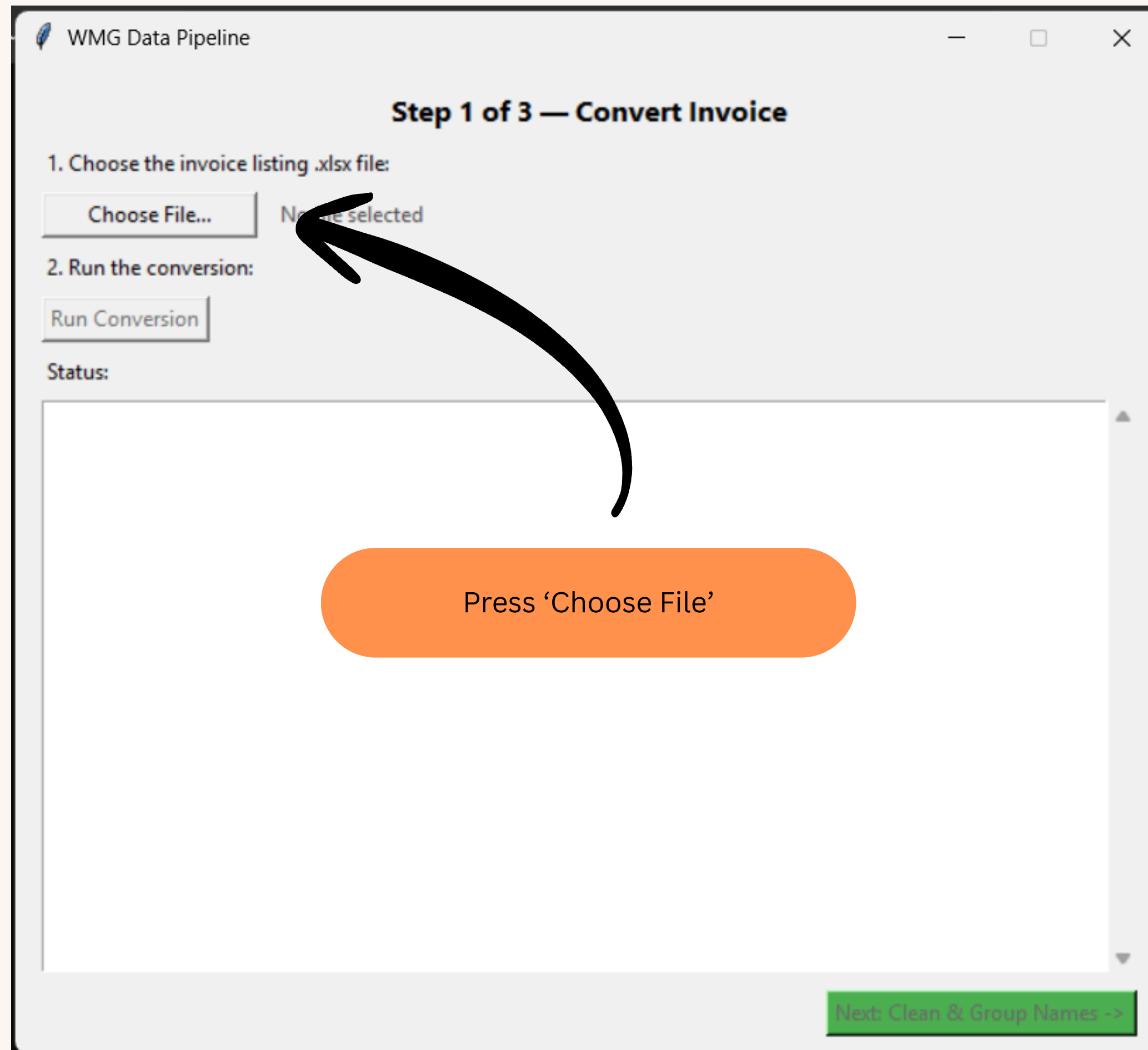
Double click and run the application



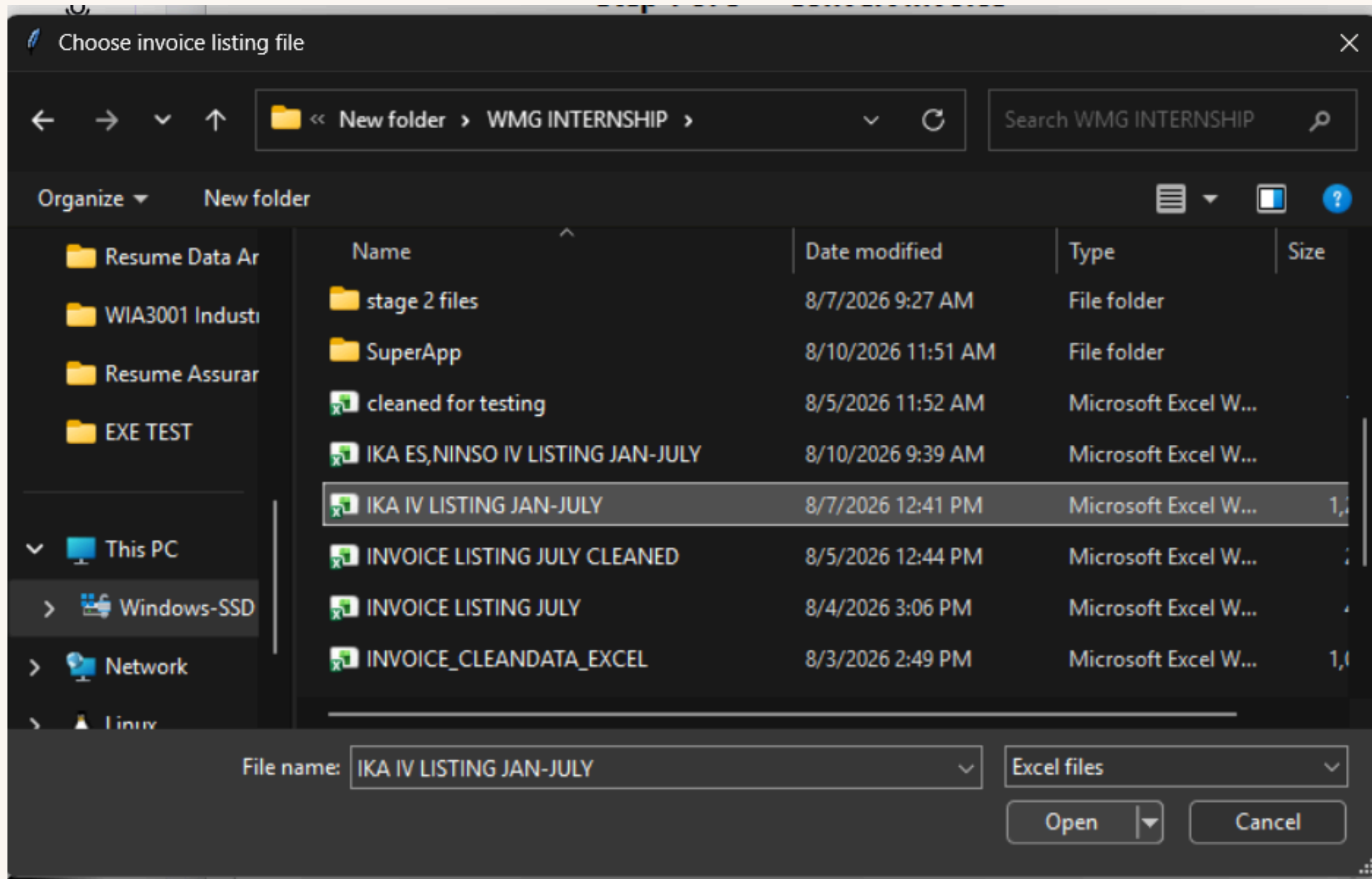
After the first run, you will notice that mapping.csv is created in the same folder
This mapping.csv is used to map the groupings, you can directly edit inside the .csv file



This is how it should look like when you open the app



This is how it should look like when you open the app



Select the 'raw' Invoice file

WMG Data Pipeline

Step 1 of 3 — Convert Invoice

1. Choose the invoice listing .xlsx file:

IKA IV LISTING JAN-JULY.xlsx

2. Run the conversion:

Status:

Selected: C:/New folder/WMG INTERNSHIP/IKA IV LISTING JAN-JULY.xlsx

Click 'Run Conversion'
it should take around 20-30 seconds

Next: Clean & Group Names ->

Step 1 of 3 — Convert Invoice

1. Choose the invoice listing .xlsx file:

Choose File...

IKA IV LISTING JAN-JULY.xlsx

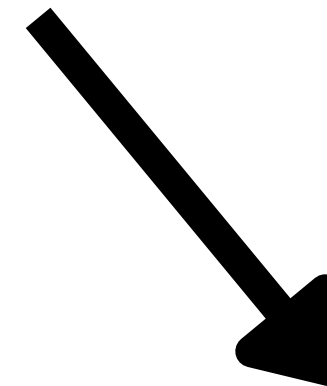
2. Run the conversion:

Run Conversion

Status:

```
Selected: C:/New folder/WMG INTERNSHIP/IKA IV LISTING JAN-JULY.xlsx  
Running conversion...  
Done. 11274 line items across 1683 invoices.  
All rows recognized, all invoice totals reconciled. Looks clean.
```

Click 'Next: Clean & Group Names'



Next: Clean & Group Names ->

Step 2 of 3 — Clean & Group Names

Received 11274 rows from Step 1.

Run Cleaning & Grouping

Manage Customer Names...

Status

Click 'Run Cleaning & Grouping'

<- Back

Next: Analyze ->

Step 2 of 3 — Clean & Group Names

Received 11274 rows from Step 1.

Run Cleaning & Grouping

Manage Customer Names...

Status

Cleaning complete. 11274 rows processed.
All names matched the mapping.

If it says “All names matched the mapping”, you are
clear to proceed to analyze

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Next: Analyze ->

Step 2 of 3 — Clean & Group Names

Received 11274 rows from Step 1.

Run Cleaning & Grouping

Manage Customer Names...

Status

Cleaning complete. 11274 rows processed.
All names matched the mapping.
Customer mapping updated and reloaded.
Note: re-run cleaning to apply the updated mapping.
Cleaning complete. 11274 rows processed.
1 name(s) not in the mapping (grouped as 'Other'):
- AEON BIG

If not, click on 'Manage Customer Names...'

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Next: Analyze ->

Manage Customer Names	
Raw name (as it appears in your data)	Group (canonical customer)
AEON (KL RDC)	AEON
BILLION MART (TANJUNG CHAT) SDN. BHD.	BILLION MART (TANJUNG CHAT) SDN. BHD.
BORONG DIN AS CASH & CARRY (ALOR JANGGUS)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY (PADANG BESAR)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY (SERIAB)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SB - PENDANG	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (JEJAWI)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (KANGAR KAPITOL)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (KANGAR)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (KUALA PERLIS)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (PAUH)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (POKOK SENA)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (SIMPANG EMPAT)	BORONG DIN AS CASH & CARRY
PASARAYA BORONG DIN AS CASH & CARRY SDN BHD (KUALA NERANG)	BORONG DIN AS CASH & CARRY
PASARAYA BORONG DIN AS SD BHD (AYER HITAM)	BORONG DIN AS CASH & CARRY
CB FROZEN FOOD SDN BHD - DC	CB FROZEN FOOD SDN BHD - DC
CS BROTHERS SDN BHD	CS BROTHERS
CS GROCER SDN BHD (KAJANG REKO)	CS GROCER
CS GROCER SDN BHD (PUCHONG)	CS GROCER

File: C:\Users\User\Desktop\mapping.csv

A new pop-up window will appear, this is where you can add/edit the grouping once u edited it, click 'Save & Close'

Step 2 of 3 — Clean & Group Names

Received 11274 rows from Step 1.

Run Cleaning & Grouping

Manage Customer Names...

Status

```
Cleaning complete. 11274 rows processed.  
All names matched the mapping.  
Customer mapping updated and reloaded.  
Note: re-run cleaning to apply the updated mapping.  
Cleaning complete. 11274 rows processed.  
1 name(s) not in the mapping (grouped as 'Other'):  
  - AEON BIG  
Customer mapping updated and reloaded.  
Note: re-run cleaning to apply the updated mapping.
```

Click on 'Run Cleaning & Grouping' one more time
now you are ready to analyze and generate the
report

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Next: Analyze ->

Step 3 of 3 — Analyze**Step 3: Analyze**

Received 11274 rows from Step 2.

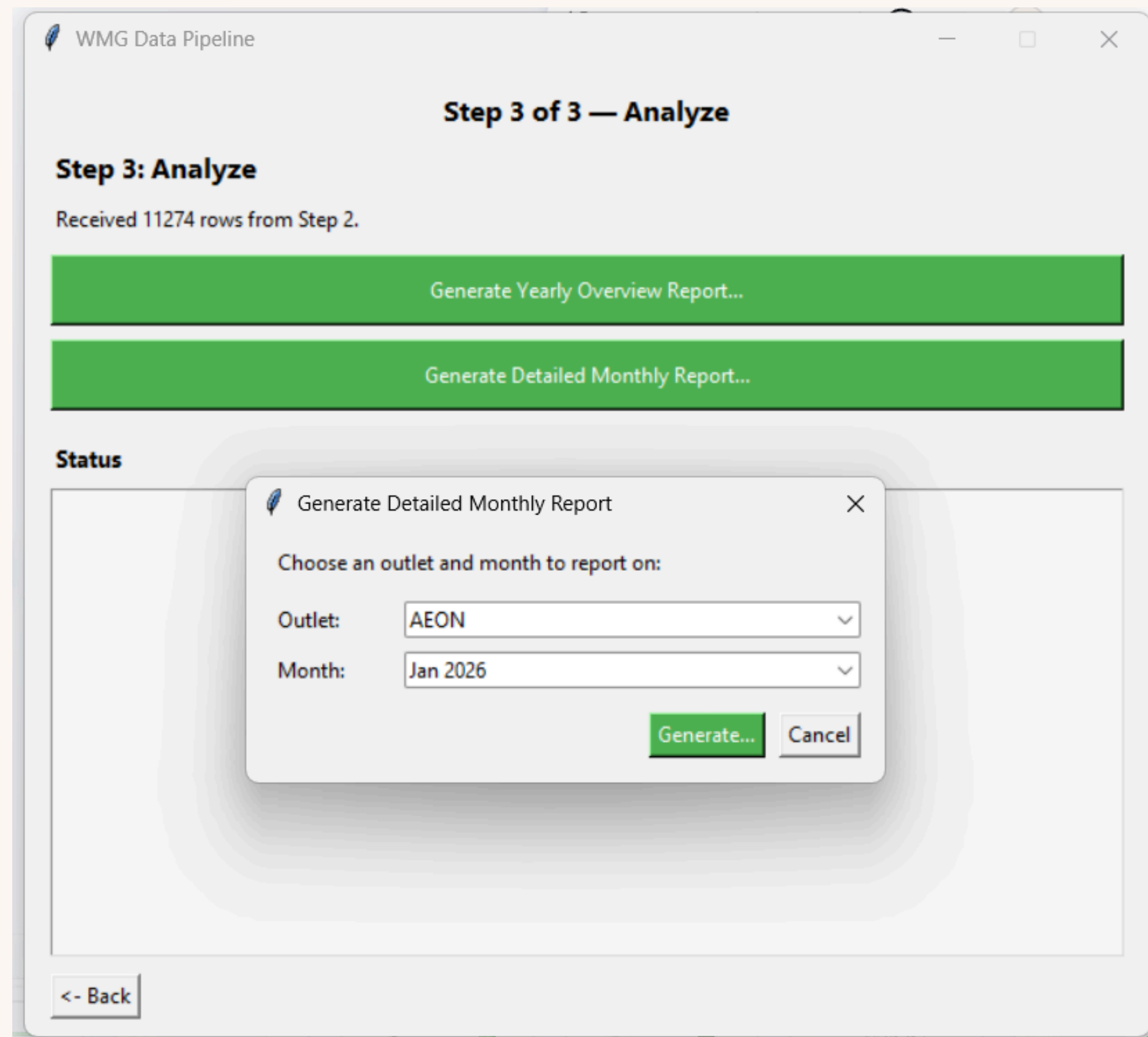
Generate Yearly Overview Report...

Generate Detailed Monthly Report...

Status

here, you will have 2 options, 'Generate Yearly Overview Report..' and 'Generate Detailed Monthly Report...'

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in ‘Generate Monthly Report..’ You will have the option to choose which month and which outlet

Finally, you save the report and you can directly view it as PDF file